NAME: ______________________ PERIOD: ________ DATE: ____________

When Is the Mean Nearly Normal?

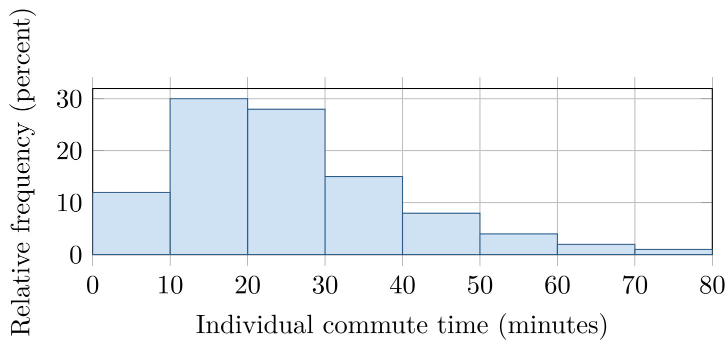
AP Statistics · Unit 2 · Topic 2.12 · B: 2026-11-13 · E: 2026-12-09

[STAR] TODAY'S PROBLEM

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: "Any average is approximately normal, even for $n = 4$."

Sol: "With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible."



FIRST TAKE — before the video (5 min)

For which sample size, $n = 4$ or $n = 40$, would you trust a normal model for the sample mean? Name one feature behind your choice.

[BOOK] CLT: SHAPE, SIZE, AND THE RIGHT DISTRIBUTION (keep on the page)

The sampling distribution of $\bar{x}$ contains the sample means from all possible random samples of one size n. For independent observations, $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$. The **central limit theorem** says that when n is sufficiently large, the sampling distribution of $\bar{x}$ is approximately normal even if the population is not. Stronger population skew generally requires a larger n; a small sample can retain that skew. The CLT concerns sample means, not individual observations.

Finish after the video:

DOK 1 **(a)** Identify the population shape and state the population mean and standard deviation with units.

DOK 2 **(b)** For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.

DOK 3 **(c)** Adjudicate both claims for $n = 4$ and $n = 40$. Cite the histogram shape, independence, and the CLT's sample-size requirement; use the $n = 40$ standard deviation; and explain one consequence of ignoring skew when using a normal model at $n = 4$.

> **Frame:** For $n = 4$, the normal model is ____________ because ________________________________; for $n = 40$, it is ____________ because ________________________________.

> **Frame:** At $n = 40$, $\mu_{\bar{x}} =$ __________ and $\sigma_{\bar{x}} =$ __________ minutes; ignoring skew at $n = 4$ could distort ____________________________.

EXIT — turn this in

One thing the video changed about my first take: ____